2025 MIT Science Bowl High School Invitational

Round 7

TOSS-UP

1) BIOLOGY – *Multiple Choice* Which of the following sources of DNA would be MOST useful for comparing distantly related species of invertebrates?

W) Mitochondrial DNA
X) Hemoglobin alpha gene
Y) lac repressor gene
Z) MADS box gene

ANSWER: W) MITOCHONDRIAL DNA

BONUS

1) BIOLOGY – *Short Answer* In chickens, dwarfism is a Z-linked recessive disease. What is the probability that a daughter of an unaffected female chicken and an affected male chicken is affected?

ANSWER: 100% (ACCEPT: 1)

TOSS-UP

2) CHEMISTRY – *Short Answer* Identify all of the following three properties of a redox reaction that are intensive: 1) Standard Gibbs free energy change of reaction; 2) Gibbs free energy change; 3) Cell potential.

ANSWER: 1 AND 3

BONUS

2) CHEMISTRY – *Short Answer* Twenty moles each of hydrogen and iodine gas are reacted in a 500 liter container. At equilibrium, eight moles of hydrogen iodide gas are present. To two significant figures, what is the reaction's equilibrium constant?

ANSWER: 0.25

TOSS-UP

3) ENERGY – *Short Answer* Scientists at the Picower Institute have been studying neurotransmitter release in neurons of Alzheimer's patients. Before being released into the synaptic cleft, neurotransmitters are found in synaptic vesicles docked to the plasma membrane via what group of water soluble proteins?

ANSWER: SNARE

BONUS

3) ENERGY – *Multiple Choice* Chemists at the Suess Group studied protein affinity for iron-sulfur clusters. Which of the following mutations in a protein's primary structure would most greatly reduce its binding to iron-sulfur clusters?

W) Cysteine (read: *Cis-teen*) to alanine
X) Threonine to isoleucine
Y) Leucine to histidine
Z) Tyrosine to phenylalanine (read: *FEEN-il-ALA-neen*)

ANSWER: W) CYSTEINE (READ: *CIS-TEEN*) TO ALANINE

TOSS-UP

4) EARTH AND SPACE – *Multiple Choice* Which of the following Milankovitch (read: *muh-LAHNG-kuh-vitch*) cycles is the shortest on Earth?

W) Axial tilt
X) Axial precession
Y) Eccentricity
Z) Apsidal precession

ANSWER: X) AXIAL PRECESSION

BONUS

4) EARTH AND SPACE – *Multiple Choice* Which of the following minerals is a framework silicate?

W) Olivine (read: *olive-een*)
X) Quartz
Y) Muscovite
Z) Amphibole

ANSWER: X) QUARTZ

TOSS-UP

5) MATH – *Short Answer* If Caroline gets a good night of sleep, then she has a 50% chance of falling asleep in lecture, but if she pulls an all-nighter, she has a 90% chance of falling asleep. If Caroline has a 20% chance of pulling an all nighter tonight, what is the probability that she falls asleep in lecture tomorrow?

ANSWER: 58% (ACCEPT: 0.58, 29/50)

BONUS

5) MATH – *Short Answer* Two concentric circles O_1 and O_2 are drawn with O_1 having a smaller radius. Chord AB of circle O_2 is drawn such that AB is tangent to O_1. Given that $AB = 12$, what is the difference between the areas of the two circles?

ANSWER: 36π

TOSS-UP

6) PHYSICS – *Multiple Choice* On a plot of temperature versus entropy, which of the following thermodynamic processes corresponds to a vertical line?

W) Isothermal
X) Adiabatic
Y) Isobaric
Z) Isochoric

ANSWER: X) ADIABATIC

BONUS

6) PHYSICS – *Short Answer* To the nearest whole number, what is the coefficient of performance of an ideal Carnot refrigerator operating with a hot reservoir at 298 kelvin and an internal temperature of 273 kelvin?

ANSWER: 11

TOSS-UP

7) BIOLOGY – *Short Answer* In order to oppose the action of the renin angiotensin aldosterone (read: *al-DOSS-tuh-rohn*) system, the atria of the heart release what hormone to inhibit Na+ reabsorption?

ANSWER: ATRIAL NATRIURETIC FACTOR (ACCEPT: ATRIAL NATRIURETIC PEPTIDE, ANP, ANF, ATRIAL NATRIURETIC HORMONE)

BONUS

7) BIOLOGY – *Short Answer* Identify all of the following three behaviors that are found in palatable species: 1) Mullerian mimicry; 2) Batesian (read: *BAYT-see-uhn*) mimicry; 3) Cryptic coloration.

ANSWER: 2 AND 3

TOSS-UP

8) CHEMISTRY – *Multiple Choice* Which of the following molecular geometries for a coordination complex with non-chiral (read: *non-KAI-ral*) ligands (read: *lig-unds*) could give both optical and geometric isomers?

W) Linear
X) Tetrahedral
Y) Square planar
Z) Octahedral

ANSWER: Z) OCTAHEDRAL

BONUS

8) CHEMISTRY – *Short Answer* Identify all of the following three statements that are true about tetrafluoroethylene: 1) It displays cis-trans isomerism; 2) It has a higher boiling point than ethylene; 3) It is highly soluble in water.

ANSWER: 2 ONLY

TOSS-UP

9) PHYSICS – *Short Answer* Ryan has a proximity sensor that detects the presence of a magnetic field by determining if a current in the device induces a voltage perpendicular to the direction of current. What effect does the sensor exploit?

ANSWER: HALL EFFECT

BONUS

9) PHYSICS – *Multiple Choice* For the time-dependent position $x(t)$ (read: *x of t*), which of the following ordinary differential equations of $x(t)$ describes a harmonic oscillator?

W) $x' + 3x = 0$ (read: *x prime plus three times x equals zero*)
X) $x'' + 3x = 0$ (read: *x double prime plus three times x equals zero*)
Y) $x' - 3x = 0$ (read: *x prime minus three times x equals zero*)
Z) $x'' - 3x = 0$ (read: *x double prime minus three times x equals zero*)

ANSWER: X) $x'' + 3x = 0$ (READ: *X DOUBLE PRIME PLUS THREE TIMES X EQUALS ZERO*)

TOSS-UP

10) ENERGY – *Short Answer* MIT physicists working on the Large Hadron Collider smashed lead ions at high enough speeds to break color confinement and create what unstable state of matter?

ANSWER: QUARK GLUON PLASMA (ACCEPT: QGP, QUARK SOUP, DO NOT ACCEPT: PLASMA)

BONUS

10) ENERGY – *Multiple Choice* Scientists at the LeBeau Group recently modeled the structure of strontium titanate using molecular dynamics simulations. Which of the following minerals has the most similar structure to strontium titanate?

W) Rutile (read: *ROO-teel*)
X) Magnetite
Y) Goethite (read: *GUR-tite*)
Z) Perovskite (read: *purr-RAWV-skyte*)

ANSWER: Z) PEROVSKITE (READ: *PURR-RAWV-SKYTE*)

TOSS-UP

11) EARTH AND SPACE – *Multiple Choice* Which of the following geologic features forms under compressional stress?

W) Normal fault
X) Horst
Y) Syncline
Z) Rift valley

ANSWER: Y) SYNCLINE

BONUS

11) EARTH AND SPACE – *Short Answer* What term describes the diapiric (read: *dye-uh-PEER-ic*) structures made primarily of evaporites which are heavily exploited in the Gulf of Mexico as they act as petroleum traps?

ANSWER: SALT DOMES

TOSS-UP

12) MATH – *Multiple Choice* An unfair coin has a probability p of landing heads on each flip. If the coin is flipped three times, the probability of landing three heads is the same as the probability of landing two heads. Which of the following is the value of p?

W) 5/8
X) 2/3
Y) 3/4
Z) 5/6

ANSWER: Y) 3/4

BONUS

12) MATH – *Short Answer* Stella is hauling her suitcase along Vassar Street in the middle of the night. Her usual walking speed is one meter per second, but two percent of the distance that she has to walk is covered in snow. Assuming that Stella can drag her suitcase through snow at 0.5 meters per second, what is her average speed over her journey, expressed in meters per second and expressed as a fraction in simplest form?

ANSWER: 50/51

TOSS-UP

13) CHEMISTRY – *Short Answer* When 1-amino-4-chloro butane (read: *one uh-MEE-no four klor-oh-BYOO-tayn*) is used as a monomer in condensation polymerization, what gas is released as a toxic byproduct?

ANSWER: HYDROGEN CHLORIDE

BONUS

13) CHEMISTRY – *Short Answer* Aaron dissolves 10 grams of a mystery solute in a 1-liter aqueous solution. Rank the following three identities of the mystery solute by increasing vapor pressure of Aaron's solution: 1) Glucose; 2) Barium chloride; 3) Sodium fluoride.

ANSWER: 3, 2, 1

TOSS-UP

14) PHYSICS – *Short Answer* A 2 kilogram ball collides and sticks with a 4 kilogram ball at rest. What fraction of the kinetic energy of the system of two balls is lost during the collision?

ANSWER: 2/3

BONUS

14) PHYSICS – *Multiple Choice* A uniformly dense galaxy is shaped like a thin disk of radius R. By what factor is the orbital period of a star in a circular orbit of radius R greater than that of a star in a circular orbit of radius $R/2$ (read: *R over two*)?

W) $1/2$ (read: *one over two*)
X) $\sqrt{2}/2$ (read: *the square root of two over two*)
Y) $\sqrt{2}$ (read: *the square root of two*)
Z) 2

ANSWER: Y) $\sqrt{2}$ (READ: *THE SQUARE ROOT OF TWO*)

TOSS-UP

15) EARTH AND SPACE – *Multiple Choice* The main sequence turnoff point of Omega Centauri, the Milky Way's most massive globular cluster, has which Harvard spectral classification?

W) O
X) A
Y) G
Z) M

ANSWER: Y) G

BONUS

15) EARTH AND SPACE – *Multiple Choice* Kordylewski clouds are objects found at the L4 Lagrange point of the Earth and what other object?

W) Venus
X) Jupiter
Y) Saturn
Z) The Moon

ANSWER: Z) THE MOON

TOSS-UP

16) MATH – *Multiple Choice* Which of the following best describes the plot of the equation $y = 1/x^2$ (read: *y equals 1 over the quantity x squared*) on a log-log scale?

W) A line with negative slope
X) A line with positive slope
Y) A parabola
Z) An exponential decay curve

ANSWER: W) A LINE WITH NEGATIVE SLOPE

BONUS

16) MATH – *Short Answer* What is the minimum positive value of a such that the graph $x + y = a$ does NOT pass through the region $x^2 + y^2 < 2025$ (read: *x squared plus y squared less than 2025*)?

ANSWER: $45\sqrt{2}$

TOSS-UP

17) BIOLOGY – *Short Answer* Brown adipose tissue is responsible for nonshivering thermogenesis. What protein in brown adipose tissue, which serves as a proton leak channel, is responsible for this generation of heat?

ANSWER: THERMOGENIN (ACCEPT: UCP1)

BONUS

17) BIOLOGY – *Short Answer* Theresia is doing a Gram stain on her sample of bacteria. Assuming she is not decolorizing using ethanol, order the following three reagents in chronological order of administration: 1) Safranin; 2) Crystal violet; 3) Iodine.

ANSWER: 2, 3, 1

TOSS-UP

18) ENERGY – *Short Answer* Scientists from the Li Lab discovered neutrons occasionally stick to quantum dots instead of scattering as expected. What fundamental interaction is responsible for this phenomenon?

ANSWER: STRONG FORCE (ACCEPT: STRONG NUCLEAR FORCE, STRONG)

BONUS

18) ENERGY – *Short Answer* Scientists at the Olsen Group are studying biopolymers containing amide bonds. Identify all of the following three synthetic polymers that also contain amide bonds: 1) Rubber; 2) Nylon; 3) Teflon.

ANSWER: 2 ONLY

TOSS-UP

19) BIOLOGY – *Multiple Choice* Which of the following pairs of amino acids can you NOT distinguish using SDS page?

W) Leucine and isoleucine
X) Selenocysteine (read: *seleno-SISS-teen*) and cysteine (read: *SISS-teen*)
Y) Pyrolysine and lysine
Z) Tyrosine and Phenylalanine

ANSWER: W) LEUCINE AND ISOLEUCINE

BONUS

19) BIOLOGY – *Short Answer* The Magic School Bus is stuck in a plasma B cell! They attach onto a protein marked with a KDEL sequence. Assuming they start in the Golgi, order the following three proteins they will meet from first to last: 1) Chaperonin (read: *shap-err-row-nin*); 2) COPI (read: *COP one*); 3) COPII (read: *COP two*).

ANSWER: 2, 1, 3

TOSS-UP

20) PHYSICS – *Multiple Choice* Elle throws a rock at an end of a uniformly dense rod attached to a pivot at its center, causing the rod to start rotating with angular velocity ω (read: *omega*). Elle then throws the same rock at the same speed towards an end of another rod of the same uniform density, but this time the rod is attached to a pivot at its other end instead of the center. In terms of ω, what is the angular velocity of Elle's new rod immediately after the collision?

W) $\omega/4$
X) $\omega/2$
Y) ω
Z) 2ω

ANSWER: X) $\omega/2$

BONUS

20) PHYSICS – *Short Answer* Nathaniel has a flexible, long rope that is sliding at a constant speed of 5 meters per second on a frictionless plane. The front end of the rope inelastically collides with a wall, causing the rope to begin accumulating in a pile at the base of the wall. If the rope has a uniform linear mass density of 2 kilograms per meter, what is the force, in Newtons, that the wall exerts on the rope during the collision?

ANSWER: 50

TOSS-UP

21) ENERGY – *Short Answer* The Broderick Group is studying parametric distributions, such as the normal distribution. How many parameters are required to completely specify a 1 dimensional normal distribution?

ANSWER: 2

BONUS

21) ENERGY – *Short Answer* Material scientists at MIT used machine learning to improve the efficiency of searching for high-entropy alloys, which contain two or more elements in equal proportions. How many distinct high-entropy alloys can be made from 5 different elements?

ANSWER: 26

TOSS-UP

22) EARTH AND SPACE – *Multiple Choice* Which of the following star types would most likely produce a type Ib (read: *one bee*) supernova?

W) Wolf-Rayet (read: *wolf RAY-ett*)
X) Carbon
Y) Cepheid (read: *SEE-fee-id*)
Z) Red Supergiant

ANSWER: W) WOLF-RAYET (READ: *WOLF RAY-ETT*)

BONUS

22) EARTH AND SPACE – *Short Answer* A space-based telescope with negligible dark current and read noise takes an image with exposure time ten thousand seconds and limiting magnitude 10. If the exposure time is increased to one million seconds, what is the new limiting magnitude?

ANSWER: 12.5

TOSS-UP

23) MATH – *Short Answer* Five students take an exam and all receive positive integer scores. Given that the unique mode of their scores is 5, what is the smallest possible mean of their scores?

ANSWER: 16/5 (ACCEPT: 3.2)

BONUS

23) MATH – *Short Answer* The radius of a sphere is growing at a constant rate of 2 centimeters per second. In cubic centimeters per second, at what rate is the volume of the sphere growing when the surface area of the sphere is 400π square centimeters?

ANSWER: 800π

TOSS-UP

24) CHEMISTRY – *Multiple Choice* Which of the following elements has the closest first ionization energy to boron?

W) Carbon
X) Magnesium
Y) Aluminum
Z) Silicon

ANSWER: Z) SILICON

BONUS

24) CHEMISTRY – *Multiple Choice* How many of carbon dioxide's vibrational modes preserve the molecule's linear geometry?

W) 0
X) 1
Y) 2
Z) 3

ANSWER: Y) 2

TOSS-UP

25) PHYSICS – *Short Answer* A conducting sphere can be considered a capacitor, with the other plate being a concentric conducting shell infinitely far away from the conductor. By what factor does the capacitance of this capacitor change if the sphere's charge and radius are both doubled?

ANSWER: 2

BONUS

25) PHYSICS – *Short Answer* Steph is watching as a mourning dove lands on the tip of a branch, causing the branch to undergo simple harmonic motion. The equilibrium position of the branch after the mourning dove lands on it is 0.1 meters below its original equilibrium position. To one significant figure and expressed in seconds, what is the period of motion of the branch?

ANSWER: 0.6